

Dialect Classification Project Report

1. Introduction

Dialect classification is a crucial task in speech processing that involves identifying the regional dialect of a speaker based on audio recordings. This project focuses on balancing an imbalanced dialect dataset and training a machine learning model for accurate classification. The dataset consists of audio recordings labeled with the speaker's **Native State**, which serves as the target variable. Since the dataset is highly imbalanced, **both undersampling and oversampling techniques** have been applied to improve model performance. Additionally, **salient segment extraction** has been performed to ensure uniformity in audio duration, enhancing feature extraction.

1.1 Dialect Detection using DASHENG Model

The project utilizes **DASHENG ((Deep Audio-Signal Holistic Embeddings) Transformer based model** for dialect detection. These models are well-suited for processing speech data and extracting meaningful linguistic patterns for classification.

1.2 Overview of Dataset

NISP-A Multilingual Multi-accent Dataset for Speaker Profiling

This dataset contains speech recordings along with speaker physical parameters (height, weight, etc.), as well as regional and linguistic information.

- **Total Speakers:** 345 (219 male, 126 female)
- **Content:** Sentences taken from newspapers
- **Duration per Speaker:** 4-5 minutes
- **Languages:** English and their respective mother tongues
- **Transcript Format:** UTF-8

For this project, we use a **subpart of this dataset** consisting of **7,839 audio files** in **Hindi and English** with variable durations ranging from **3 to 33 seconds**. The key classification target is the **Native State** of the speakers.

2. Methodology

The methodology consists of **dataset preprocessing, data balancing, feature extraction, model training, and evaluation**. The dataset includes metadata such as **Speaker ID, Gender, Mother Tongue, Native State, and District**. Since some dialects are underrepresented, balancing techniques such as **undersampling** and **data augmentation** are applied.

2.1 Preparation of the Dataset

- First, the **NISP-Dataset** repository ([iiscleap/NISP-Dataset](https://github.com/iiscleap/NISP-Dataset)) was cloned and saved locally.
- All audio files were consolidated into a single folder named “**Merged Audio**”.
- A **pandas DataFrame** was created containing the **file paths, Speaker IDs, and corresponding Native States**.

2.2 Data Preprocessing and Balancing the Dataset

Since the dataset is highly imbalanced, preprocessing steps are necessary to **balance the dataset** before training.

Balancing the Dataset

Undersampling Majority Classes:

- Classes with more than **350 samples** are **randomly undersampled** to **retain only 350 samples**.

Oversampling Minority Classes:

- Classes with fewer than **350 samples** undergo **data augmentation** to increase their count.
- Augmentation techniques include:
 - **Time Stretching** (modifying speed while preserving pitch)
 - **Pitch Shifting** (altering pitch without affecting tempo)

- **Noise Addition** (introducing slight noise for variation)
- Each original file is augmented **up to three times** to bridge the gap between available and target counts.
- The newly generated augmented files are added back to the dataset, ensuring a **balanced distribution** across different Native States.

In the Oversampling, we have used **Vocal Tract Length Perbutation (VTLP)** technique to oversample the audio data. To achieve **vocal tract normalization**, the frequency axis of the spectrogram of each speaker is linearly warped using a **warp factor** , that accounts for the relative length of their vocal tract compared to the cannonical mean.

2.3 Extracting Salient Segments and Feature Extraction

Since the audio duration ranges upto 33 seconds, extracting key 3-second segments from audio files longer than 3 seconds is crucial to ensuring a uniform input length and enabling the model to process the data effectively. Instead of padding shorter audio samples, we **repeated them** to match the highest duration in the dataset.

To extract key audio segments, we implemented the `extract_salient_segments` function, which:

- Computes **spectral flux**, identifying regions of maximum spectral change.
- Detects **peak points** in the spectral flux, ensuring high-energy segments are captured.
- Extracts **3-second segments** from these peak points, ensuring a uniform and meaningful feature representation.

While passing audio files to the Dasheng feature extractor, the maximum duration is set to **3 seconds** for effective data processing. In the second case, the audio files inherently had a maximum duration of 3 seconds. In the third case, after data augmentation, some audio files extended up to **41 seconds**, but only the first 3 seconds were considered when extracting features for the DASHENG Transformer model.

3. Implementation Details

3.1 Augmentation

- VTLP (Vocal Tract Length Perturbation) is applied to underrepresented class samples by warping the frequency axis of their spectrograms using warp factors between 0.9 and 1.1.
- Multiple augmented variants are generated for each minority class sample to increase class diversity and reduce bias.
- Augmented samples are treated as regular data—segmented into 3-second clips and passed through the same feature extraction pipeline.

3.2 Feature Extraction

- Initially, **salient segments** are extracted from each audio file using spectral flux analysis.
- Features are then extracted using **DashengFeatureExtractor**, which processes audio inputs with a maximum duration of 3 seconds for effective feature extraction.
- The dataset is converted into a **DatasetDict** format for efficient processing.

3.3 Model Training

- The processed dataset is used to train a dialect classification model.
- The model is fine-tuned using the pre-extracted features.
- Training is conducted using a **train-validation split (80%-20%)**.

3.4 Evaluation Metrics

- Model performance is evaluated using:

- **Accuracy**
- **F1-score**
- **Precision & Recall**
- **Confusion Matrix**

4. Results & Discussion

Several experiments were conducted to analyze the impact of different preprocessing techniques on the model's performance. The following four cases were evaluated:

1. Neural Networks on Extracted 3-Second Audio Data with MFCC Features:

- **Approach:** In this case, instead of using the DASHENG feature extractor, 3-second audio segments were processed to extract **MFCC (Mel-Frequency Cepstral Coefficients)** features, which were then fed into a neural network classifier.
- **Result:** The model achieved a notably lower **accuracy of 0.2041**.
- **Discussion:** This result suggests that the neural network configuration and the MFCC feature extraction approach were not as effective for this task as the DASHENG Transformer-based method. The low accuracy indicates that MFCCs, while useful in many speech tasks, might not capture the necessary dialect-specific features as effectively as the transformer-based representations in this context.

2. Extracting 3-Second Audio Segments Without Balancing:

- **Approach:** The dataset remained imbalanced, and only the salient **3-second segments** were extracted from each audio file. The Dasheng

feature extractor was applied with a max duration of 3 seconds.

- **Result:** The model achieved an **accuracy of 0.684**.
- **Discussion:** The low accuracy indicates that the uneven class distributions negatively impacted classification performance despite the uniform segment length.

3. **Balancing the Dataset Without Extracting 3-Second Audio:**

- **Approach:** Balancing was applied using undersampling and oversampling, but full-length audio files were used instead of extracting 3-second segments. Notably, after augmentation, some audio files reached up to 41 seconds, yet only the first 3 seconds were considered during feature extraction.
- **Result:** The model achieved an **accuracy of 0.736**.
- **Discussion:** Balancing improved performance by reducing bias. However, the inclusion of longer audio files, even though truncated, could introduce variability in the extracted features.

4. **Balancing the Dataset with Different Augmentation techniques and Extracting 3-Second Audio Segments:**

- **Approach:** Both balancing techniques and salient segment extraction were combined, ensuring that each audio input was a consistent 3-second clip fed into the Dasheng feature extractor.
- **Result:** The model achieved an **accuracy of 0.861**.
- **Discussion:** This combination proved the second most effective. Balancing addressed class bias while extracting consistent, high-information segments improved feature quality, leading to significant performance gains.

5 . **Balancing the Dataset with VTLP and Extracting 3-Second Audio Segments:**

- **Approach:** Data balancing using Vocal Tract Length Perturbation (VTLP) and consistent 3-second audio segment extraction. VTLP simulates variations in speaker vocal tract length by warping the frequency axis of audio spectrograms using a warp factor (typically between 0.9 and 1.1). This generates new, realistic variations of existing audio samples without altering their linguistic content.
- **Result:** This method achieved the highest accuracy of **.8885** across all experiments.
- **Discussion:** The combination of VTLP for data augmentation and standardized segment extraction proved to be the most effective. VTLP helped reduce bias by augmenting minority classes with realistic variations, while fixed-length segments ensured consistent and meaningful input. Together, these strategies enhanced model generalization and feature quality, resulting in the best overall performance.

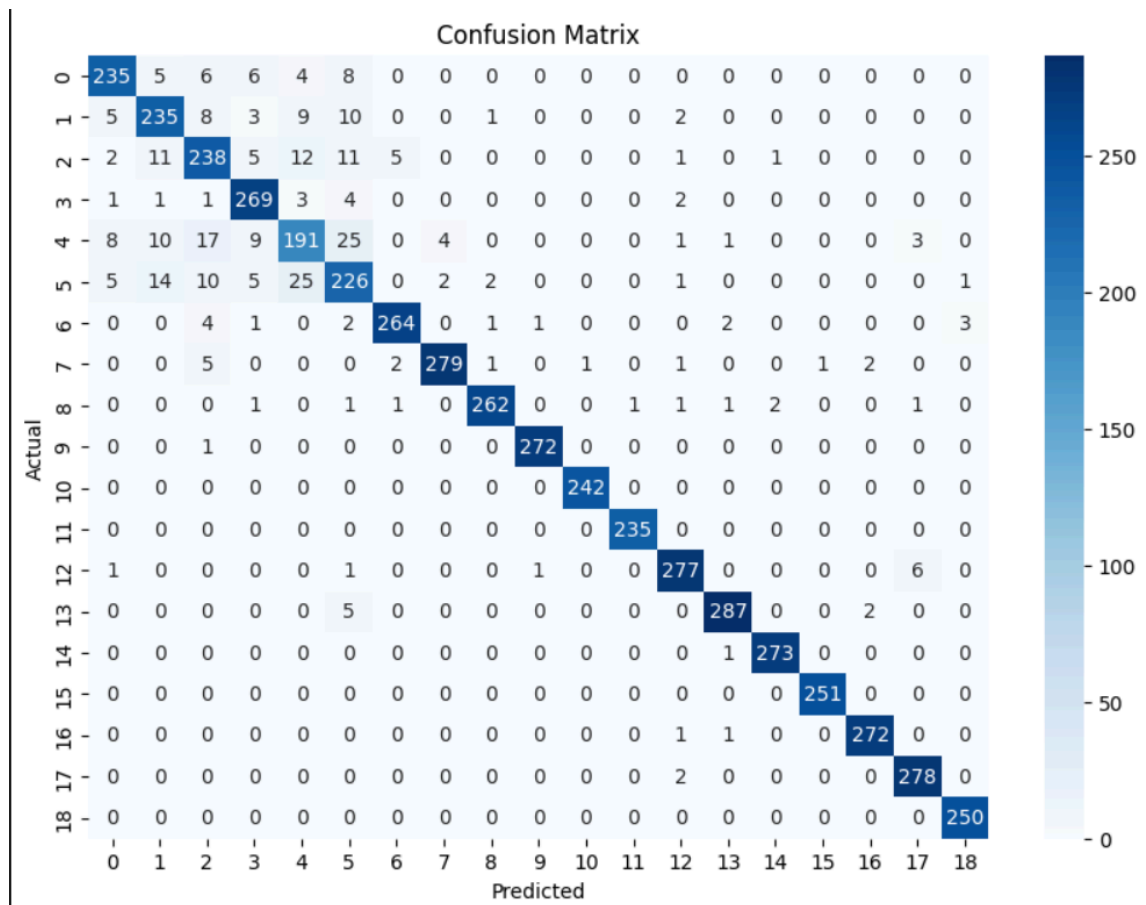
Overall Effect of Different Approaches:

- **Extracting salient 3-second segments alone** improved feature uniformity, but the imbalanced classes limited performance.
- **Balancing the dataset alone** helped reduce bias, yet longer audio files (even if truncated) introduced unnecessary variability.
- **Combining balancing with 3-second extraction** resulted in the best performance by ensuring both balanced class distributions and uniform, high-quality feature representations.
- **Using MFCC-based features with neural networks** performed poorly compared to transformer-based methods, highlighting the importance of choosing the right feature extraction approach for dialect classification.

Table 1: Comparison of the evaluation metrics for the Dasheng Transformer model under two configurations

Metric	Without VTLP Balancing	With VTLP Balancing
Evaluation Loss	2.1597	2.1383
Evaluation Accuracy	0.8661	0.8885
Evaluation F1-score	0.8650	0.8892
Evaluation Precision	0.8679	0.8911
Evaluation Recall	0.8661	0.8885
Training Accuracy	0.9365	0.9413
Evaluation Runtime (s)	8.829	8.9406
Evaluation Samples/Second	146.336	144.509
Evaluation Steps/Second	4.644	4.586

- Confusion Matrix without VTLP



- Confusion Matrix with VTLP

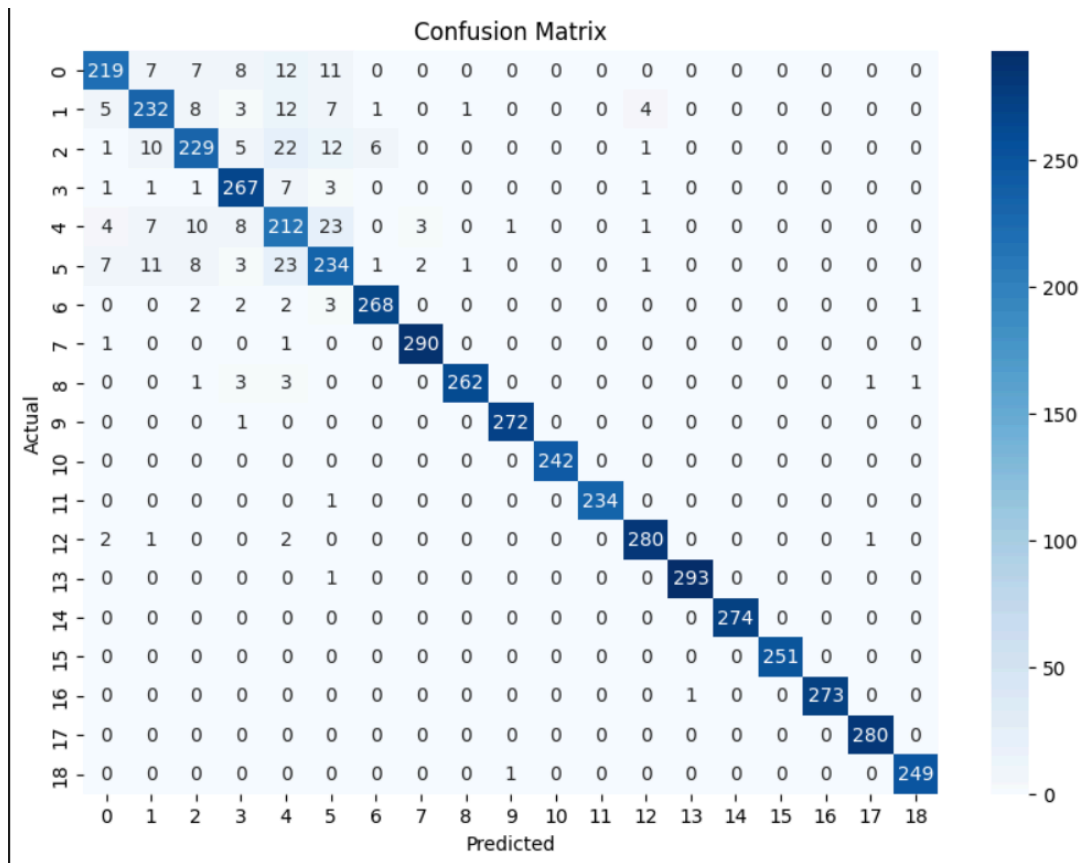


Table 2: Experiment vs Accuracy

Experiment

Accuracy

Neural Networks on 3-Second Audio with MFCC Features

0.2041

Extracting 3-Second Audio Segments Without Balancing	0.684
Balancing the Dataset Without Extracting 3-Second Audio	0.736
Balancing with Different Augmentation Techniques & Extracting 3-Second Audio	0.861
Balancing with VTLP & Extracting 3-Second Audio Segments (Best)	0.889

5. Conclusion & Future Work

- The dialect classification model effectively classifies speakers based on their native state.
- Data augmentation and balancing techniques played a crucial role in improving performance.
- The **extraction of key 3-second segments** further improved accuracy by focusing on high-information portions of the audio.
- In contrast, using MFCC features with a neural network did not yield satisfactory results, underscoring the superiority of transformer-based feature extraction for this task.

- Future work can explore **advanced augmentation techniques** and **fine tuning the hyperparameters Transformer architectures** to further enhance performance, as well as fine-tuning the neural network approach for MFCC-based models.